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Inspiring Excellence

Department of Mathematics and Natural Sciences

MAT216 : Linear Algebra and Fourier Analysis

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Linear Equation: A *linear equation* is an equation in which the highest power of the unknowns is 1 & no product of unknowns.

General Form of a System of Linear Equations: An arbitrary system of m linear equations in n unknowns can be written as:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= b_m. \end{aligned}$$

Here, $x_1, x_2, \ldots, x_n$ are the unknowns, and the subscripted a's and b's denote constants.

Augmented Matrix Form of a System of Linear Equations (SLE)

$$\left[\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{nn} & b_m \end{array}\right].$$

Pivot :- 1st non zero entry of a row of the matrix. (column) - নিচের

$$\begin{pmatrix} 1 & 2 & 3 & 7 \\ 0 & 0 & 5 & 6 \end{pmatrix}$$

Row Echelon Form (REF):

1. All zero rows (if any) are at the bottom.
2. For each row: The first **non-zero entry (pivot)** is strictly to the **right** of the first **non-zero entry** of the **row above**.
3. In row-echelon form all entries below any pivot must be zero.

Reduced Row Echelon Form (RREF):

1. **The matrix** is in **Row Echelon** Form.
2. All pivot element must be 1.
3. All entries above pivot must be zero.

Idea of Gaussian Elimination Method (REF):

- ☐ **Swapping two equations (rows)** doesn't change the system.

- ☐ **Multiplying an equation (row) by a non-zero constant** doesn't change the system.
- ☐ **Adding or subtracting two equations (rows)** and replacing one of them with the resulting equation doesn't change the system.

Idea of Gauss-Jordan Elimination Method (RREF):

- ☐ **The Augmented matrix** is in **Row Echelon** Form.
- ☐ All pivot element must be 1.
- ☐ All entries above pivot must be zero.

Homework

Use Gaussian Elimination Method (equivalent to Row Echelon Form)/ Gauss-Jordan elimination (equivalent to Reduced Row Echelon Form) to solve the following system of linear equations.

$$\begin{aligned} 2x-y+z&=-3\\ 4x-y+z&=-5\\ x+3y-2z&=4 \end{aligned}$$

$$\begin{aligned} x-4y+5z&=8\\ 3x+7y-z&=3\\ x+15y-11z&=-14 \end{aligned}$$

$$\begin{aligned} 2x-3y+5z&=1\\ 3x+y-z&=2\\ x+4y-6z&=1 \end{aligned}$$

$$\begin{aligned} -2y+3z&=1\\ 3x+6y-3z&=-2\\ 6x+6y+3z&=5 \end{aligned}$$

$$\begin{aligned} y+2z &= 3\\ 2x+4y-3z &= -4\\ 3x+5y-5z &= -6 \end{aligned}$$

$$\begin{aligned} x-y+2z-w &= -1\\ 2x+y-2z-2w &= -2\\ -x+2y-4z+w &= 1\\ 3x-3w &= -3 \end{aligned}$$

$$\begin{aligned} -3x_2+2x_3+x_4 &= -1\\ x_1+2x_3-10x_4 &= 10\\ -x_1+x_2-2x_3+7x_4 &= -7 \end{aligned}$$

$$\begin{aligned} x_1-x_2-x_3+2x_4 &= 1\\ 2x_1-2x_2-x_3-3x_4 &= 3\\ -x_1+x_2-x_3 &= -3 \end{aligned}$$

$$\begin{aligned} x_1+3x_2+2x_3+2x_5&=0\\ 2x_1+6x_2-5x_3-2x_4+4x_5-3x_6&=-1\\ 5x_3+10x_4+15x_6&=5\\ 2x_1+6x_2+8x_4+4x_5+18x_6&=6 \end{aligned}$$

$$\begin{aligned} 10x_2-4x_3+x_4&=1\\ x_1+4x_2-x_3+x_4&=2\\ 3x_1+2x_2+x_3+2x_4&=5\\ -2x_1-8x_2+2x_3-2x_4&=-4\\ x_1-6x_2+3x_3&=1 \end{aligned}$$

$$\begin{aligned} 2x_1-x_2+3x_3+4x_4&=9\\ x_1-2x_3+7x_4&=11\\ 3x_1-3x_2+x_3+5x_4&=8\\ 2x_1+x_2+4x_3+4x_4&=10 \end{aligned}$$

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Equivalent Matrices: Two matrices A and B are said to be **equivalent** if one can be obtained from the other by a finite sequence of **elementary row / elementary column operations.**

Example: Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad R_2 \to R_2 - 3R_1, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}.$$

Hence, $A \sim B$ and therefore A and B are **equivalent matrices**.

Identity Matrix:: A $n \times n$ identity matrix is a square matrix with 1 on the main diagonal and 0 elsewhere. It is often denoted by I.

Examples: $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Inverse of a Matrix:
The square matrix A^{-1} is called the inverse of a square matrix A if $A \cdot A^{-1} = A^{-1}A = I$.

Permutation Matrix:: A permutation matrix is a square matrix obtained by performing a row or column swap on an identity matrix.

Example: Swapping the two rows of $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ gives $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

Elementary Matrix:: An elementary matrix is a square matrix obtained by performing an elementary row or column operation on an identity matrix.

Example: Adding twice the first row to the second row of $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ gives $E = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$.

Elementary Row Operations:

1. Swapping any two rows. $[R_i \leftrightarrow R_j]$ $(R_i \leftrightarrow R_j)$
2. Multiplying a row by a non-zero constant. $[R_i' = kR_i]$ $(R_i' = k \cdot R_i$, where $k \neq 0)$
3. Adding or subtracting a multiple of one row to another row. $(R_i' = R_i \pm k \cdot R_j)$

$[R_i' = R_i \pm kR_j]$

Homework

1. What is Elementary Matrix and Permutation Matrix?

2. Write down 3 random Elementary Matrix and Permutation Matrix.

3. Find the inverse of $A = \begin{bmatrix} 1 & 4 \\ 3 & 0 \end{bmatrix}$ using the Gauss–Jordan elimination method (reduced row echelon form). (Canonical form). (normal form).

4. Find A^{-1} for $A = \begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$ by using the Gauss–Jordan elimination method (reduced row echelon form).

5. Consider $A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & 4 & 7 \\ 3 & 7 & 5 \end{bmatrix}$. Justify the statement "A is not invertible" using row operations.

6. (a) Calculate the inverse of the matrix using Gauss–Jordan elimination method:

$$A = \begin{bmatrix} 2 & 2 & 3 \\ 1 & -3 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

(b) Hence solve the system of equations:

$$\begin{cases} 2x + 2y + 3z = 1 \\ x - 3y + z = 2 \\ x + 2y = 5 \end{cases}$$

7. Solve the following system of equations using inverse matrix technique. (Calculate the inverse matrix using row operations).

$$\begin{aligned} x_1 + 3x_2 + 2x_3 &= 5 \\ 2x_1 + x_2 + 3x_3 &= 1 \\ 3x_1 + 2x_2 + z &= 4 \end{aligned} \qquad \begin{aligned} 2x_1 - x_2 - x_3 &= 6 \\ x_1 + 3x_2 + 2x_3 &= 1 \\ 3x_1 - x_2 - 5x_3 &= 1 \end{aligned}$$

8. Determine the values of λ and μ such that the system $x+y+z=6$, $x+2y+3z=10$, $x+2y+\lambda z=\mu$ has (i) a unique solution, (ii) more than one solution, and (iii) no solution.

9. Determine the values of δ such that the system $x+y-z=1$, $2x+3y+\delta z=3$, $x+\delta y+3z=2$ has (i) a unique solution, (ii) more than one solution, and (iii) no solution.

10. **Question:** Determine the values of a and b such that the system represented by the augmented matrix $\left[\begin{array}{ccc|c} a & 0 & b & 2 \\ a & a & 4 & 4 \\ 0 & a & 2 & b \end{array}\right]$ has (i) a unique solution, (ii) more than one solution, and (iii) no solution.

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Mathematical Structure : A mathematical structure is a non-empty set or collection of objects that is equipped with one or more operations, relations or laws that satisfy specific rules or axioms.

Examples: Group, Ring, Field, Vector Space.

Binary Operation:: A binary operation $*$ on a set S is a mapping

$$* : S \times S \to S.$$

Suppose $*$ is a binary operation. A set G is called *closed under* $*$ if

$$a * b \in G \quad \forall\, a, b \in G.$$

Note: all binary operations satisfy the closure property by definition.

Examples:

1. Addition on the set $\mathbb{R}$, operation $\mathbf{a} * \mathbf{b} = \mathbf{a} + \mathbf{b}$.
2. Subtraction on the set $\mathbb{R}$, operation $\mathbf{a} * \mathbf{b} = \mathbf{a} - \mathbf{b}$.
3. Addition on the set $\mathbb{Z}$, operation $\mathbf{a} * \mathbf{b} = \mathbf{a} + \mathbf{b}$.

Group:: A nonempty set G is said to be a group if in G there is defined an operation $*$ such that:

- **Closure:** $a * b \in G$, for all $a, b \in G$.
- **Associativity:** $a * (b * c) = (a * b) * c$, for all $a, b, c \in G$.
- **Existence of Identity:** There exists a special element $e \in G$ such that $e * a = a * e = a$, for all $a \in G$.
- **Existence of Inverse:** For every $a \in G$, there exists an element $a' \in G$ such that $a * a' = a' * a = e$, for all $a \in G$.

Examples:

1. The set of all integers $\mathbb{Z}$ under the binary operation of addition.
2. The set $\mathbb{R}$ under the binary operation of addition.
3. The set $\mathbb{R}^{\times} = \mathbb{R} - \{0\}$ under the binary operation of multiplication.

Field:: A non-empty set F is called a **Field** under the binary operations $\oplus$ and $\otimes$ if:

- F is an Abelian group under $\oplus$, and e_0 is the identity element of the operation $\oplus$.
- $F^{\times} = F - \{e_0\}$ is an Abelian group under the operation $\otimes$.
- There is a connection called **Distributive law** between $\oplus$ and $\otimes$ such that

$$a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c).$$

Examples:

1. The set $\mathbb{R}$ under the binary operations of addition and multiplication.
2. The set $\mathbb{C}$ under the binary operations of addition and multiplication.
3. The set $\mathbb{Q}$ under the binary operations of addition and multiplication.

Homework (Vector Space):

21. Check whether the set $V = \mathbb{R}$ (set of real numbers) with the usual addition and scalar multiplication is a vector space or not:
    - The set $V = \mathbb{R}$ over the field $F = \mathbb{R}$ (set of real numbers)
    - The set $V = \mathbb{R}$ over the field $F = \mathbb{Q}$ (set of rational numbers)
    - The set $V = \mathbb{R}$ over the field $F = \mathbb{C}$ (set of complex numbers)
22. Show that the set of complex numbers $\mathbb{C}$ is a vector space over $\mathbb{R}$.
23. Show that the set of real numbers $\mathbb{R}$ is a vector space over the field $\mathbb{Q}$.
24. Show that the set of all 3×3 matrices of real entries form a vector space under usual operations over the field $\mathbb{R}$.
25. Show that the set of all 3×3 matrices of complex entries form a vector space under usual operations over the field $\mathbb{C}$.
26. Show that the set of all 3×3 matrices of complex entries form a vector space under usual operations over the field $\mathbb{R}$.
27. Show that the set of all 3×3 matrices of real entries **do not form** a vector space under usual operations over the field $\mathbb{C}$.
28. In computer graphics, a 2D screen is denoted by the set $V = \{(x, y, 1) : x, y \in \mathbb{R}\}$. The operations of addition and scalar multiplication are defined on V by $(x_1, y_1, 1) + (x_2, y_2, 1) = (x_1 + x_2, y_1 + y_2, 1)$ & $k(x_1, y_1, 1) = (kx_1, ky_1, 1)$. Is the set V with these operations a vector space over the field $\mathbb{R}$?
29. Let V=$\mathbb{R}^+$ be the set of all positive real numbers. We define vector addition & scaler multiplication by $x \oplus y = xy \; \forall x, y \in \mathbb{R}^+$ & $k \circ x = x^k \; \forall x \in \mathbb{R}^+$ & $k \in \mathbb{R}$. Is V a vector space over the field $\mathbb{R}$? Justify your answer.
30. Let V=$\mathbb{R}^2$ be the The set of all pairs of real numbers of the form (x, y), where $x \geq 0$; with the standard operations on $\mathbb{R}^2$. Is V a vector space over the field $\mathbb{R}$? Justify your answer.

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Definition of Vector Space:

A set **V** equipped with two binary operations **addition & scaler multiplication** *is called a* ***vector space*** over the field **F** if **V** satisfies the following **10 axioms**.

- **(A1) Closed under Addition:** $\vec{u}+\vec{v} \in V$ for all $\vec{u},\vec{v} \in V$.
- **(A2) Commutativity:** $\vec{u}+\vec{v}=\vec{v}+\vec{u} \in V$ for all $\vec{u},\vec{v} \in V$.
- **(A3) Associativity:** $(\vec{u}+\vec{v})+\vec{w}=\vec{u}+(\vec{v}+\vec{w}) \in V$ for all $\vec{u},\vec{v},\vec{w} \in V$.
- **(A4) Additive Identity:** There exists a $\vec{0} \in V$ such that $\vec{u}+\vec{0}=\vec{u}$ for all $\vec{u} \in V$.
- **(A5) Additive Inverse:** There exists a $\vec{u}' \in V$ such that $\vec{u}+\vec{u}'=\vec{0}$ for all $\vec{u} \in V$.
- **(M1) Closed under Scaler Multiplication:** $\alpha\vec{u} \in V$ for all $\alpha \in F$ & $\vec{u} \in V$.
- **(M2) Distributive Law of Vectors:** $\alpha(\vec{u}+\vec{v})=\alpha\vec{u}+\alpha\vec{v}$ for all $\alpha \in F$ & $\vec{u},\vec{v} \in V$.
- **(M3) Distributive Law of Scalers:** $(\alpha+\beta)\vec{u}=\alpha\vec{u}+\beta\vec{u}$ for all $\alpha,\beta \in F$ & $\vec{u} \in V$.
- **(M4) Associativity of Scaler Multiplication:** $(\alpha\beta)\vec{u}=\alpha(\beta\vec{u})$ for all $\alpha,\beta \in F$ & $\vec{u} \in V$.
- **(M5) Unit of Scaler Multiplication:** $1\vec{u}=\vec{u}$ where $1 \in F$ & $\vec{u} \in V$.

Homework

1. Consider the set $V=\mathbb{R}^2$ with addition $(x_1,y_1)+(x_2,y_2)=(x_1+x_2,y_1+y_2)$ and scalar multiplication $k(x_1,y_1)=(kx_1,0)$. Is $\mathbb{R}^2$ with these operations a vector space over $\mathbb{R}$? M5 X

2. Let $V=\mathbb{R}^2$ denote the set of all ordered pairs of real numbers. If (a_1,a_2) and (b_1,b_2) are elements of V and $c \in \mathbb{R}$, define vector addition and scalar multiplication by $(a_1,a_2)+(b_1,b_2)=(a_1+b_1,a_2b_2)$ and $c(a_1,a_2)=(ca_1,a_2)$. Is V a vector space over $\mathbb{R}$ with respect to these operations? Justify your answer. A5X

3. Let $V=\mathbb{R}^2$ denote the set of all ordered pairs of real numbers. If (a_1,a_2) and (b_1,b_2) are elements of V and $c \in \mathbb{R}$, define vector addition and scalar multiplication by $(a_1,a_2)+(b_1,b_2)=(a_1+b_1,a_2+b_2)$ and $c(a_1,a_2)=(ca_1,a_2)$. Is V a vector space over $\mathbb{R}$ with respect to these operations? Justify your answer. M3 X

4. Consider the set $V=\mathbb{R}^2$. A generic element of $\mathbb{R}^2$ is given by the pair (x,y), where $x,y \in \mathbb{R}$ and $k \in \mathbb{R}$. The operations addition and scalar multiplication are defined on $\mathbb{R}^2$ by $(x_1,y_1)+(x_2,y_2)=(x_1+x_2+1,y_1+y_2+1)$ and $k(x_1,y_1)=(kx_1,ky_1)$. Is the set $\mathbb{R}^2$ with these operations a vector space over the field $\mathbb{R}$?

5. Consider the set $V = \mathbb{R}^2$. A generic element of $\mathbb{R}^2$ is given by the pair (x,y), where $x,y \in \mathbb{R}$ and $k \in \mathbb{R}$. The operations addition and scalar multiplication are defined on $\mathbb{R}^2$ by $(x_1,y_1)+(x_2,y_2) = (x_1+x_2, 3y_1y_2)$ and $k(x,y) = (k^2x, 0)$. Is the set $\mathbb{R}^2$ with these operations a vector space over the field $\mathbb{R}$?

6. Consider the set $V = \mathbb{R}^3$. A generic element of $\mathbb{R}^3$ is given by the triple (x,y,z), where $x,y,z \in \mathbb{R}$ and $k \in \mathbb{R}$. The operations addition and scalar multiplication are defined on $\mathbb{R}^3$ by $(x,y,z)+(x',y',z') = (x+x', y+y', z+z')$ and $k(x,y,z) = (k^2x, k^2y, k^2z)$. Is the set $\mathbb{R}^3$ with these operations a vector space over the field $\mathbb{R}$?

7. Consider the set $V = \mathbb{R}^3$. A generic element of $\mathbb{R}^3$ is given by the triple (x,y,z), where $x,y,z \in \mathbb{R}$ and $k \in \mathbb{R}$. The operations addition and scalar multiplication are defined on $\mathbb{R}^3$ by $(x,y,z)+(x',y',z') = (x+x', y+y', z+z')$ and $k(x,y,z) = (0,0,0)$. Is the set $\mathbb{R}^3$ with these operations a vector space over the field $\mathbb{R}$?

8. Show that all the 3-ordered tuples of real numbers will make a vector space under usual operations over the field $\mathbb{R}$. Or Show that $\mathbb{R}^3$ is a vector space over the field $\mathbb{R}$.

9. Let $V = \{(0,0,0)\}$ & $V_1 = \{(1,1,1)\}$. Are V & V_1 vector spaces over $\mathbb{R}$ with respect to the usual operations? Justify your answer.

10. Show that the set of all 2×2 matrices of real entries form a vector space under usual operations over the field $\mathbb{R}$.

11. Show that the set of all $m\times n$ matrices of real entries form a vector space under usual operations over the field $\mathbb{R}$.

12. Show that the set of all 2×2 invertible real matrices with the standard matrix addition and multiplication does not form a vector space over the field $\mathbb{R}$.

13. Show that the set of all matrices with real entries of the form $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, with the standard matrix addition and multiplication, forms a vector space over the field $\mathbb{R}$.

14. Show that the set of all matrices with real entries of the form $\begin{pmatrix} a & 1 \\ 1 & b \end{pmatrix}$, with the standard matrix addition and multiplication does not form a vector space over the field $\mathbb{R}$.

15. Show that the set of all matrices with real entries of the form $\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix}$ with the matrix addition and scaler multiplication defined by $\begin{bmatrix} a_1 & 1 \\ 1 & b_1 \end{bmatrix} + \begin{bmatrix} a_2 & 1 \\ 1 & b_2 \end{bmatrix} = \begin{bmatrix} a_1+a_2 & 1 \\ 1 & b_1+b_2 \end{bmatrix}$ & $k\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix} = \begin{bmatrix} ka & 1 \\ 1 & kb \end{bmatrix}$ form a vector space over the field $\mathbb{R}$.

16. The set of all functions/ continuous functions/ differentiable functions/ Integrable functions $f : \mathbb{R} \to \mathbb{R}$ forms a vector space over the field $\mathbb{R}$.

17. The set of all discontinuous functions $f : \mathbb{R} \to \mathbb{R}$ forms a vector space over the field $\mathbb{R}$.

18. The set of all functions $f : \mathbb{R} \to \mathbb{R}$ with $f(3) = 0$ forms a vector space over the field $\mathbb{R}$ but $f : \mathbb{R} \to \mathbb{R}$ with $f(3) = 5$ doesnot form a vector space over the field $\mathbb{R}$.

19. All Polynomials of degree ≤ 3 with real coefficients form a vector space over the field $\mathbb{R}$. The set of all Polynomials of degree $= 3$ with real coefficients do not form a vector space over the field $\mathbb{R}$.

20. Does the set of integers $\mathbb{Z}$ form a vector space over the field $\mathbb{R}$?

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Subspace : Let V be a vector space. A subset W of V is called a subspace of V if W itself is a vector space under the same operations and field.

Testing for Subspaces: A subset $\mathbf{W}$ of a vector space $\mathbf{V}$ is called a subspace of $\mathbf{V}$ if and only if

1. (A4) $\vec{0} \in \mathbf{W}$
2. (A1) For any $\vec{u}, \vec{v} \in \mathbf{W} \Rightarrow \vec{u} + \vec{v} \in \mathbf{W}$
3. (M1) For any scalar α and any $\vec{u} \in \mathbf{W} \Rightarrow \alpha\vec{u} \in \mathbf{W}$

$\alpha\vec{u} + \beta\vec{v} \in V$

The set $\mathbb{R}^n$ forms a vector space over the field $\mathbb{R}$ under the usual vector addition and scalar multiplication.

The set of all $m \times n$ matrices with real entries, denoted by $M_{m\times n}(\mathbb{R})$, forms a vector space over $\mathbb{R}$ under the usual operations.

The set of all $m \times n$ matrices with **complex** entries, denoted by $M_{m\times n}(\mathbb{R})$, forms a vector space over $\mathbb{R}/\mathbb{C}$ under the usual operations.

The set of all functions $f : \mathbb{R} \to \mathbb{R}$ (or $f : [a,b] \to \mathbb{R}$) forms a vector space over the field $\mathbb{R}$ under the usual addition and scalar multiplication.

Homework:

1. Determine whether the set $S = \{(a,0,0) : a \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$ or not. satisfies হ্যাঁ
2. Determine whether the set $S = \{(a,b,c) : a,b,c \in \mathbb{R} \text{ and } a+b+c = 3\}$ is a subspace of $\mathbb{R}^3$ or not.
3. Show that W is a subspace of $\mathbb{R}^3$. W is the set of all vectors of the form (a,b,c) with $a,b,c \in \mathbb{R}$ and $a+b-c = 0$. হ্যাঁ $\to \{\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ for t
4. satisfies Is the set of vectors $W = \{(x,y,z) : y = -4x - z, z = 8x\}$ a subspace of the vector space $V = \mathbb{R}^3$?
5. All vectors of the form (x,y,z) with $x + 2y + 3z = 5$, where the total vector space $V = \mathbb{R}^3$ over the scalar field $F = \mathbb{R}$, with usual addition and scalar multiplication. not
6. Let W be the subset of a vector space $V = \mathbb{R}^3$ defined as $W = \{(r,0,s) : r^2 + s^2 = 0, r,s \in \mathbb{R}\}$. Determine whether W is a subspace of V.

A4 ✓
A1 ✓

Check whether the set of vectors $S = \{\vec{x} \in \mathbb{R}^n : A \cdot \vec{x} = 0\}$ is a subspace of the vector space $\mathbb{R}^n$, where A is an $m \times n$ matrix.

8. The set $M_{2\times 2}(\mathbb{R})$ is the set of all 2×2 matrices with real entries that form a vector space over $\mathbb{R}$. Determine whether the set $S = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a,b,c,d \in \mathbb{R} \text{ and } a+b+c+d = 0 \right\}$ is a subspace of $M_{2\times 2}(\mathbb{R})$ or not.

9. The set $M_{2\times 2}(\mathbb{R})$ is the set of all 2×2 matrices with real entries that form a vector space over $\mathbb{R}$. Determine whether the set $S = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} : a,b \in \mathbb{R} \right\}$ is a subspace of $M_{2\times 2}(\mathbb{R})$ or not.

10. Consider the subset W of all 3×3 invertible matrices from $M_{3\times 3}(\mathbb{R})$. Will this subset be a subspace of $M_{3\times 3}(\mathbb{R})$?

11. Let $M_{3\times 3}(\mathbb{R})$ be the vector space of all 3×3 real matrices, and let $W = \{A \in M_{3\times 3}(\mathbb{R}) \mid \det(A) = 0\}$; determine whether W is a subspace of $M_{3\times 3}(\mathbb{R})$.

12. Consider the subset $W = \{A \in M_{3\times 3}(\mathbb{R}) \mid A^\top = A\}$ of all 3×3 real symmetric matrices; determine whether W is a subspace of $M_{3\times 3}(\mathbb{R})$ over the field $\mathbb{R}$.

13. Consider the subset $W = \{A \in M_{3\times 3}(\mathbb{R}) \mid A^\top = -A\}$ of all 3×3 real skew-symmetric matrices; determine whether W is a subspace of $M_{3\times 3}(\mathbb{R})$ over the field $\mathbb{R}$.

14. Consider the subset $W = \{A \in M_{3\times 3}(\mathbb{C}) \mid A^\dagger = A\}$ of all 3×3 Hermitian matrices; determine whether W is a subspace of $M_{3\times 3}(\mathbb{C})$ over the field $\mathbb{R}$. ✓

15. Consider the subset $W = \{A \in M_{3\times 3}(\mathbb{C}) \mid A^\dagger = A\}$ of all 3×3 Hermitian matrices; determine whether W is a subspace of $M_{3\times 3}(\mathbb{C})$ over the field $\mathbb{C}$. ×

16. Consider the subset $W = \{A \in M_{3\times 3}(\mathbb{C}) \mid A^\dagger = -A\}$ of all 3×3 skew-Hermitian matrices; determine whether W is a subspace of $M_{3\times 3}(\mathbb{C})$ over the field $\mathbb{R}$.

17. Consider the subset $W = \{A \in M_{3\times 3}(\mathbb{C}) \mid A^\dagger = -A\}$ of all 3×3 skew-Hermitian matrices; determine whether W is a subspace of $M_{3\times 3}(\mathbb{C})$ over the field $\mathbb{C}$.

18. The set of all functions $f : \mathbb{R} \to \mathbb{R}$ with $f(3) = 0$ forms a subspace over the field $\mathbb{R}$.

19. The set of all functions $f : \mathbb{R} \to \mathbb{R}$ with $f(3) = 5$ do not form a subspace over the field $\mathbb{R}$.

20. The set of all Continuous functions $f : \mathbb{R} \to \mathbb{R}$ form a subspace over the field $\mathbb{R}$.

21. The set of all discontinuous functions $f : \mathbb{R} \to \mathbb{R}$ do not form a subspace over the field $\mathbb{R}$.

22. The set of all differentiable functions $f : \mathbb{R} \to \mathbb{R}$ form a subspace over the field $\mathbb{R}$.

23. All Polynomials of degree ≤ 3 with real coefficients form a subspace over the field $\mathbb{R}$.

24. All Polynomials of degree = 3 with real coefficients do not form a subspace over the field $\mathbb{R}$.

6

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MAT216 : Linear Algebra and Fourier Analysis

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Linear Combination: Let V be a vector space over the field F. If $v_1, v_2, v_3, \ldots, v_n \in V$ and $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n \in F$, then

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \cdots + \alpha_n v_n$$

is called a **linear combination.**

Linear Independence: A set of vectors $\{v_1, v_2, v_3, \ldots, v_n\}$ is called a **linearly independent set** if the homogeneous system

$$\alpha_1 \vec{v_1} + \alpha_2 \vec{v_2} + \alpha_3 \vec{v_3} + \cdots + \alpha_n \vec{v_n} = 0$$ unique sol^n → independent

has only the zero (unique) solution. In other words, all $\alpha_i = 0$. $(0,0,0) = (0)$

Linear Dependence: A set of vectors $\{v_1, v_2, v_3, \ldots, v_n\}$ is called a **linearly dependent set** if the homogeneous system

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \cdots + \alpha_n v_n = 0$$

has **infinitely many solutions**. In other words, not all α_i are zero. infinite many sol^n → dependent.

Dependence for two vectors:

- A finite set of vectors that contains the zero vector is **linearly dependent**.
- A set with exactly two vectors is **linearly independent** if and only if neither vector is a scalar multiple of the other.

Dependence for more vectors: A set S with two or more vectors is

- **Linearly dependent** if and only if at least one of the vectors in S is expressible as a linear combination of the other vectors in S.
- **Linearly independent** if and only if no vector in S is expressible as a linear combination of the other vectors in S.

Linear Dependence in R^n

Let $S = \{v_1, v_2, v_3, \ldots, v_k\}$ be a set of vectors in $\boxed{\mathbb{R}^n}$. If $\boxed{k > n}$, then S is linearly dependent.
Note: The converse is not true.

$k > n$ → dependent

$k \leq n$ → no conclusion

Homework

$\boxed{\alpha_1 = 2,\ \alpha_2 = 3,\ \alpha_3 = 1}$

(1) In the vector space $M_{2\times 2}(\mathbb{R})$, express the vector $A = \begin{bmatrix} 5 & 1 \\ -2 & 3 \end{bmatrix}$ as a linear combination of the vectors

$$A_1 = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix},\ A_2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix},\ \text{and } A_3 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

$A = \alpha_1 A_1 + \alpha_2 A_2 + \alpha_3 A_3$

[last sem mid] $\alpha_1 = 2$, $\alpha_2 = 3$, $\alpha_3 = 1$ | $\alpha_i \in \mathbb{R}$, $i = \overline{1,3}$

(2) In the vector space $\mathbb{R}^3$, express the vector $\mathbf{v} = (6, -7, 3)$ as a linear combination of the vectors $\mathbf{v}_1 = (1, -2, 1)$, $\mathbf{v}_2 = (2, 1, -1)$, and $\mathbf{v}_3 = (7, -4, 1)$. $(\alpha_1, \alpha_2, \alpha_3) = (4-3t,\ 1-2t,\ t)$

(3) In the vector space $\mathbb{R}^3$, the vector $\mathbf{v} = (1, 2, 6)$ cannot be expressed as a linear combination of the vectors $\mathbf{v}_1 = (2, 1, 0)$, $\mathbf{v}_2 = (1, -1, 2)$, and $\mathbf{v}_3 = (0, 3, -4)$. → No soln

(4) Show that the set of vectors $\{(2, 1, 2), (0, 1, -1), (4, 3, 4)\}$ in $\mathbb{R}^3$ is linearly independent. → unique soln

(5) Show that the set of vectors $\{(2, 1, 2), (0, 1, -1), (4, 3, 3)\}$ in $\mathbb{R}^3$ is linearly dependent. → many soln

(6) Are the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

linearly independent?

(7) Given the set of vectors (exam 2 ปีที่ 3).

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix},\quad \mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix},\quad \mathbf{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix},$$

1. Determine whether the vectors are linearly independent or dependent.
2. If they are linearly dependent, express one of the vectors as a linear combination of the others.

(8) Check whether the following set of vectors are linearly independent or dependent:

$k = 4$, $\mathbb{R}^3 = 3$ (dim) $S = \{(-1, 5, 2), (3, 1, 1), (2, 0, -2), (4, 1, 0)\}$. $4 > 3$ → dependent

9) Show that the set of vectors $\{(3, 0, 1, -1), (2, -1, 0, 1), (1, 1, 1, -2)\}$ in $\mathbb{R}^4$ is linearly dependent.

10) Determine whether the following vectors in $\mathbb{R}^4$ are linearly independent: $\mathbf{v}_1 = (3, 3, 0, 0)$, $\mathbf{v}_2 = (0, 0, 2, 2)$, and $\mathbf{v}_3 = (1, 1, 0, -1)$.

Naqib

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Department of Mathematics and Natural Sciences

MAT216 : Linear Algebra and Fourier Analysis

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Linear Span: Let V be a vector space over a field F, and let $S = \{v_1, v_2, ..., v_n\} \subseteq V$. The linear span (or simply **span**) of S, denoted by Span(S), is defined as the set of all linear combinations of the vectors in S:

$$\text{Span}(S) = \{a_1v_1 + a_2v_2 + \cdots + a_nv_n \mid a_i \in F\}$$

Spanning Set: A set of vectors $S = \{v_1, v_2, ..., v_n\} \subseteq V$ is called a **spanning set** of V if:

$$\text{Span}(S) = V$$

no solⁿ → no span

That is, every vector in V can be written as a linear combination of vectors in S. *Mathematically, for any* arbitrary $\vec{v} \in V$, the system

$$\alpha_1\vec{v}_1 + \alpha_2\vec{v}_2 + \alpha_3\vec{v}_3 + \cdots + \alpha_k\vec{v}_k = \vec{v}$$

has a solution (unique / many).

***Note:* If the set $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\}$ is a basis of V, then the solution is unique. Otherwise, if the set is linearly dependent, the solution may not be unique.**

unique → basis

Examples in $\mathbb{R}^2$

Example 1: Spanning the plane Let $v_1 = (1,0)$, $v_2 = (0,1)$. Then,

$$\text{Span}\{v_1, v_2\} = \{(a,b) \mid a,b \in \mathbb{R}\} = \mathbb{R}^2$$

So, $\{v_1, v_2\}$ is a spanning set of $\mathbb{R}^2$.

Example 2: Line through the origin Let $v = (2,3)$. Then,

$$\text{Span}\{v\} = \{a(2,3) \mid a \in \mathbb{R}\}$$

This set forms a line through the origin in the direction of $(2,3)$.

set ↗ element span

Theorem-1

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \ldots, \mathbf{v}_k\} \subset \mathbb{R}^n$. If $k < n$, then S **cannot** span $\mathbb{R}^n$. If $k < n$, then $\text{Span}(S) \neq \mathbb{R}^n$.

Homework

1) Show that vectors (1,2), (2,4) do not span the whole $\mathbb{R}^2$.

$\alpha_1 + 2\alpha_2 = a$

$2\alpha_1 + 4\alpha_2 = b.$

$\left(\begin{array}{cc|c} a & 1 & a \\ 2 & 4 & b \end{array}\right)$

No solⁿ → not span

2) Show that vectors (1,2), (-2,4) & (3,2) span the whole $\mathbb{R}^2$.

pivot unknown → many solⁿ. → span

3) • Show that the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix}$$

unique sol^n

span the vector space $\mathbb{R}^3$.

4) • Determine whether the set of vectors S spans $\mathbb{R}^3$, where S={ (3,0,4) ,(6,2, -1), (12,4, -2) , (3,2, -5) }

5) • Let $\mathbf{v}_1 = (1,3,0,-2), \mathbf{v}_2 = (3,-6,3,5), \mathbf{v}_3 = (4,-2,1,3)$ & $\mathbf{v}_4 = (2,-6,1,3)$.
Determine whether the set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ spans $\mathbb{R}^4$.

Basis: Let V be any vector space, and let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \ldots, \mathbf{v}_k\}$ be a set of vectors in V. Then S is called a **basis** for V if the following two conditions hold:

- ☐ S is linearly independent.
- ☐ S spans V.

basis is not unique

Note: Basis ⇔ Minimal Spanning Set

Dimension: If V is a vector space and $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \ldots, \mathbf{v}_n\}$ is a basis for V, then the number of elements n is called the **dimension** of V. $\dim(V) = n$.

Theorem-2

If V is any n-dimensional vector space ($\mathbb{R}^n$) and $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \ldots, \mathbf{v}_k\}$, then:

- *If number of vectors (k) < dimension (n), then S cannot span V.*
- If number of vectors (k) > dimension (n), then S is dependent.

Theorem-3

If V is any n-dimensional vector space and $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \ldots, \mathbf{v}_n\}$ is a set of n vectors in V, then S is called a basis if: **S is linearly independent.**

Unique Representation of Vectors

If V is any vector space and $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \ldots, \mathbf{v}_k\}$ is a **basis** for V, then Any vector $\vec{v} \in V$ can be expressed as a linear combination of the vectors in S in **exactly one way (unique)**.

Homework

6) Is $S = \{(1,2),(2,3)\}$ a basis of the space of $\mathbb{R}^2$. basis

7) Is $S = \{(1,2),(2,4)\}$ a basis of the space of $\mathbb{R}^2$. not basis

8) Show that the set of vectors S do not form a basis of $\mathbb{R}^3$ $S = \{(2,-3,1),(4,1,1),(0,-7,1)\}$. dependent

9) Show that the set of vectors S form a basis of $\mathbb{R}^3$ $S = \{(2,-3,1),(4,1,1),(0,-7,2)\}$. unique sol^n → basis.

10) Show that the set of vectors S do not form a basis of $\mathbb{R}^3$ $S = \{(2,-3,1),(4,1,1)\}$. dim - 3 → not span ∴ basis

11) Show that the set of vectors S form a basis of $\mathbb{R}^3$ $S = \{(1,2,3),(4,5,6),(7,8,10)\}$.

basis = independent + span

independent → basis

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8

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MAT216 : Linear Algebra and Fourier Analysis

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Linear Transformation: Let U and V be two vector spaces over the same field $\mathbb{F}$. A linear transformation T of U into V, written as $T : U \to V$, is a function from U into V such that:

- $T(u_1 + u_2) = T(u_1) + T(u_2)$ for all $u_1, u_2 \in U$
- $T(\alpha u) = \alpha T(u)$ for all $u \in U$ and all $\alpha \in \mathbb{F}$

$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+2y \\ x-y \end{pmatrix} \Rightarrow T(x) = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$

$= x\begin{pmatrix} 1 \\ 1 \end{pmatrix} + y\begin{pmatrix} 2 \\ -1 \end{pmatrix}$

Linear Operator: Let V be a vector space over a field $\mathbb{F}$. If the transformation $T : V \to V$ is a linear transformation from V into itself, then T is called a **Linear Operator**.

Domain

Co-domain

same

Note: A Linear Transformation is determined by the movement of the basis vectors of the domain.

Matrix Representation of Linear Transformation: Let the linear transformation be defined by:

$$T(\vec{x}) = A\vec{x}$$

where $\vec{x}$ is a **column vector**, A is a **matrix**, and T is the **transformation**.

Image of a Linear Transformation: Let $T : U \to V$ be a linear transformation over the field $\mathbb{F}$. Here, U is called the **domain** and V is called the **codomain** of T. Then the **image** is the range of T, which is denoted by $Im(T)$.

Note: Image is the Column Space of the Matrix of T.

null space

Kernel of a Linear Transformation: Let $T : U \to V$ be a linear transformation over the field $\mathbb{F}$. Here, U is called the **domain** and V is called the **codomain** of T. Then the **kernel** of T is defined by

basis dimension

$$\ker T = \{u \in U \mid T(u) = 0\}$$

Note: Kernel is the Null Space of the Matrix of T.

Rank-Nullity Theorem: Let $T : U \to V$ be a linear transformation over the field $\mathbb{F}$. Then,

$$\boxed{\overset{\to Dim(Im(T))}{\text{Rank}(T)} + \overset{\to ker(T)}{\text{Nullity}(T)} = \dim(U)}$$ /No. of rows /column

One-one / Injective Linear Transformation: Let $T : U \to V$ be a linear transformation over the field $\mathbb{F}$. Here, U is called the **domain** and V is called the **codomain** of T.If

$$\ker(T) = \{0\} \quad \text{or} \quad \text{Nullity}(T) = 0,$$

then T is called a **one-one (injective)** linear transformation.

Onto / Surjective Linear Transformation: Let $T : U \to V$ be a linear transformation over the field $\mathbb{F}$. Here, U is called the **domain** and V is called the **codomain** of T. If

$$\text{Range}(T) = \text{Im}(T) = V \quad \text{or} \quad \text{Rank}(T) = \dim(V),$$

then T is called an **onto (surjective)** linear transformation.

Invertibility: Let $T : U \to V$ be a linear transformation over the field $\mathbb{F}$. T is called **invertible** if and only if T is both **one-one (injective)** and **onto (surjective)**.

Homework

1. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the transformation defined by $T(x,y,z) = (x+1,\ x+y,\ x+z)$. Show that T is a linear transformation.

2. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the transformation defined by $T(x,y,z) = (x+y-z,\ 0,\ x+2y-z)$. Show that T is a linear transformation.

3. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the transformation defined by $T(x,y,z) = (xy,\ x+y,\ x+z)$. Show that T is not a linear transformation.

4. Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be defined by $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ x+y \\ x-y \end{pmatrix}$. Show that T is a linear transformation.

5. Consider a linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ such that $T(1,0) = (-1,1)$ & $T(0,1) = (2,1)$; find $T(x,y)$ & $T(7,8)$.

6. Consider a linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^3$ such that $T\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \\ 4 \end{pmatrix}$ and $T\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$; find $T\begin{pmatrix} x \\ y \end{pmatrix}$ & $T\begin{pmatrix} 8 \\ 9 \end{pmatrix}$.

7. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation defined by $T(x,y) = (x+4y,\ 3x+2y)$. Find the matrix representation of T with respect to standard basis.

8. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation defined by $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2x-3y+z \\ x+y-4z \\ 2y+5z \end{pmatrix}$. Find the matrix representation of T with respect to standard basis.

9. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be a linear transformation defined by $T(x_1,x_2,x_3,x_4) = (3x_1-2x_2+x_4,\ x_2-x_3+x_4,\ x_1+3x_2-4x_3-2x_4)$. Find the matrix representation of T with respect to standard basis.

10. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the transformation defined by $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} |x|+2y \\ x+y \end{bmatrix}$. Is T a linear transformation?

11. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation defined by $T(x,y,z) = (2x-y+z,\ x+2y-z,\ x+7y-4z)$. Find the Basis and Dimension of Im(T) & Find the Basis and Dimension of Ker(T). Verify **Rank-Nullity** theorem.

12. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be a linear transformation defined by $T(x,y,z,w) = (x-2y-z-2w,\ -2x+4y+6z+8w,\ 3x-6y+4z+w)$. Find the Basis and Dimension of Im(T) & Find the Basis and Dimension of Ker(T). Verify **Rank-Nullity** theorem.

13. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x-3y \\ x+y \end{pmatrix}$. Show that T is invertible and find T^{-1}.